GREEN LIFE - HAVE KNOWLEDGE. PAY ATTENTION.

Green Life Screen Map

1. Primary App Structure

Bottom navigation for MVP:

- Home
- Search
- Saved
- Profile

Primary entry actions from Home:

- Scan Barcode
- Take Photo
- Search Product

Modal and secondary flows:

- Login / Sign Up
- Permissions
- Result Details
- Source Details
- Error / No Match
- Save Confirmation

2. Screen-by-Screen Map

A. Onboarding

Purpose:

Explain value and request required permissions.

Content:

- App name and value proposition
- "Scan products. Understand environmental impact."
- Short explanation of confidence labels
- CTA: Continue
- Optional CTA: Sign in later

Actions:

- Continue to permissions
- Skip to app shell

B. Permissions

Purpose:

Request camera access for scanning.

Content:

- Camera permission request
- Optional photo library permission
- Explanation of why access is needed

Actions:

- Allow camera
- Not now

C. Auth

Variants:

- Sign In
- Sign Up
- Continue as Guest

Purpose:

Allow account creation for saved history.

Content:

- Email
- Password or passwordless option
- Guest option

Actions:

- Sign in
- Create account
- Continue as guest

D. Home

Purpose:

Primary launchpad for item identification.

Content:

- Search bar
- Primary CTA: Scan Barcode
- Secondary CTA: Take Photo
- Recent items preview
- Short note: "Results may be exact or estimated"

Actions:

- Open barcode scanner
- Open camera/photo flow
- Start search
- Open recent item

E. Barcode Scanner

Purpose:

Fastest exact-match path.

Content:

- Camera viewfinder
- Manual code entry fallback
- Flash toggle
- Cancel

Actions:

- Scan code
- Enter code manually
- Cancel to Home

Outcomes:

- Exact match -> Result Details
- No match -> No Match / Search Fallback

F. Search

Purpose:

Manual lookup by product, brand, material, or category.

Content:

- Search input
- Suggested recent searches
- Search results list
- Filter chips:
- Products
- Brands
- Categories
- Materials

Actions:

- Enter query
- Select result

- Refine search

Outcomes:

- Result selected -> Result Details
- No result -> No Match

G. Photo Capture / Upload

Purpose:

Fallback for non-barcode items and future OCR/object recognition.

Content:

- Camera capture
- Upload from library
- Guidance text: "Capture label, brand, or product clearly"

Actions:

- Take photo
- Upload image
- Retake
- Continue

Outcomes:

- OCR/match found -> Candidate Results or Result Details
- Weak match -> Search suggestions
- No match -> No Match

H. Candidate Results

Purpose:

Let user disambiguate between likely matches.

Content:

- "We found possible matches"
- Ranked list of products/categories
- Confidence hint per option

Actions:

- Select a candidate
- Search again
- Cancel

I. Result Details

Purpose:

Core product value screen.

Sections:

- Item header
- product/category name
- brand if known
- image if available
- Match banner
- Exact Product Match
- Probable Product Match
- Category Estimate
- Material Estimate
- Summary metrics
- Carbon footprint
- Material composition
- Recycled content
- Recyclability
- Disposal guidance
- Hazard flags if available
- Source section
- Save button
- Compare alternatives placeholder for future phase

Actions:

- Save item

- Expand metric details
- Open source details
- Share result later if added

J. Metric Detail

Purpose:

Explain one metric in more detail.

Content:

- Metric name
- Value and unit
- Scope
- Methodology summary if available
- Source
- Dataset date/version
- Confidence note

Actions:

- Open source details
- Back to result

K. Source Details

Purpose:

Build trust through provenance.

Content:

- Source name
- Source type: EPA / EPD / manufacturer / factor dataset
- Record date or ingestion date
- Link/reference metadata
- Notes on exact vs estimated derivation

Actions:

- Back to result

L. No Match

Purpose:

Handle lookup failure gracefully.

Content:

- "We couldn't find an exact match"
- Suggested next steps:
- Try search
- Retake photo
- Scan again
- View category estimates if possible

Actions:

- Retry scan
- Search manually
- Use closest category

M. Saved

Purpose:

Retain long-term value.

Content:

- Saved items list
- Recent scans/searches
- Sort by recent / name

Actions:

- Open saved item
- Remove saved item

N. Profile

Purpose:

Account and settings.

Content:

- Account status
- Sign in/out
- Permission settings
- About confidence labels
- Privacy and terms links

Actions:

- Manage account
- Sign out
- Open help/about

O. Admin Web Panel

Internal only.

Views:

- Product search
- Source records
- Match review queue
- Confidence anomalies
- Manual mapping override

3. Core Navigation Flow

```
flowchart TD
    A["Onboarding"] --> B["Permissions"]
    B --> C["Auth or Guest"]
    C --> D["Home"]
    D --> E["Barcode Scanner"]
    D --> F["Search"]
    D --> G["Photo Capture"]
```

```
E --> I["Result Details"]
E --> L["No Match"]
F --> I
F --> L
G --> H["Candidate Results"]
G --> L
H --> I
I --> J["Metric Detail"]
I --> K["Source Details"]
I --> M["Saved"]
```

4. Text Wireframes

Home

```
-------------------------------------------------
| Green Life                                     |
| Search products, brands, materials             |
| [ Search bar............................... ]  |
|                                                |
| [ Scan Barcode ]                               |
| [ Take Photo  ]                                |
|                                                |
| Recent                                          |
| - Stainless Bottle                              |
| - Laundry Detergent                             |
| - Cardboard Box                                 |
|                                                |
| Note: Results may be exact or estimated.       |
|                                                |
| Home   Search   Saved   Profile                |
-------------------------------------------------
```

Barcode Scanner

```
-------------------------------------------------
| < Back                    Scan Barcode         |
|                                                |
|               [ Camera Viewfinder ]            |
|                                                |
| Align barcode inside the frame                 |
|                                                |
| [ Flash ]            [ Enter Code Manually ]   |
-------------------------------------------------
```

Search

```
-------------------------------------------------
| Search                                         |
| [ stainless water bottle................... ]  |
```

```
|                                                |
| Filters: [Products] [Brands] [Categories]      |
|                                                |
| Results                                         |
| ----------------------------------------------
| Reusable Water Bottle                          |
| Example Brand                                  |
| Probable Product Match                         |
| ----------------------------------------------
| Stainless Steel Drinkware                      |
| Category Estimate                              |
| ----------------------------------------------
| Vacuum Bottle                                  |
| Another Brand                                  |
| Exact Product Match                            |
| ----------------------------------------------
|                                                |
| Home   Search   Saved   Profile                |
--------------------------------------------------
```

Photo Capture / Upload

```
--------------------------------------------------
| < Back                     Add Photo           |
|                                                |
|             [ Camera Preview / Image ]         |
|                                                |
| Capture a label, product, or brand clearly     |
|                                                |
| [ Take Photo ]                                 |
| [ Upload From Library ]                        |
|                                                |
| After capture:                                 |
| [ Retake ] [ Continue ]                        |
--------------------------------------------------
```

Candidate Results

```
--------------------------------------------------
| Possible Matches                               |
| We found a few likely results                  |
|                                                |
| ----------------------------------------------
| Reusable Water Bottle                          |
| Example Brand                                  |
| Confidence: High                               |
| [ Select ]                                     |
| ----------------------------------------------
| Stainless Steel Drinkware                      |
| Category Estimate                              |
| Confidence: Medium                             |
| [ Select ]                                     |
```

```
| ------------------------------------------------
|                                                |
| [ Search Instead ]                             |
-------------------------------------------------
```

Result Details

```
-------------------------------------------------
| < Back                     Item Details        |
|                                                |
| [ Product Image ]                              |
| Reusable Water Bottle                          |
| Example Brand                                  |
|                                                |
| [ Exact Product Match ]                        |
| Confidence: 96%                                |
|                                                |
| Key Impact                                      |
| Carbon Footprint      2.4 kg CO2e              |
| Recycled Content      35%                      |
| Recyclability         Moderate                 |
| Material              Stainless Steel          |
| Disposal              Check local metal rules  |
|                                                |
| [ Save Item ]                                  |
|                                                |
| Sources                                         |
| EPD Database                                   |
| EPA Dataset                                    |
| [ View Source Details ]                        |
|                                                |
| Home   Search   Saved   Profile                |
-------------------------------------------------
```

Metric Detail

```
-------------------------------------------------
| < Back                 Carbon Footprint        |
|                                                |
| Carbon Footprint                               |
| 2.4 kg CO2e                                    |
|                                                |
| Scope: Product                                 |
| Estimate Type: Exact                           |
| Confidence: 91%                                |
|                                                |
| Methodology                                    |
| EPD-reported cradle-to-gate value              |
|                                                |
| Source                                         |
| EPD Database                                   |
| Version: 2026-01                               |
```

```
| Published: Jan 12, 2026                        |
| [ View Source ]                                |
-------------------------------------------------
```

Source Detail

```
-------------------------------------------------
| < Back                   Source Detail         |
|                                                |
| EPD Database                                   |
| Type: EPD                                      |
| Reference URL: [ link ]                        |
| Version: 2026-01                               |
| Published: Jan 12, 2026                        |
| Ingested: Feb 1, 2026                          |
|                                                |
| Notes                                          |
| Cradle-to-gate environmental declaration       |
-------------------------------------------------
```

No Match

```
-------------------------------------------------
| No Exact Match Found                           |
|                                                |
| We couldn't find this item exactly.            |
|                                                |
| Try one of these:                              |
| [ Search by name ]                             |
| [ Retake photo ]                               |
| [ Scan again ]                                 |
| [ Use category estimate ]                      |
-------------------------------------------------
```

Saved

```
-------------------------------------------------
| Saved                                          |
|                                                |
| ---------------------------------------------- |
| Reusable Water Bottle                          |
| Example Brand                                  |
| Saved Mar 5                                    |
| ---------------------------------------------- |
| Laundry Detergent                              |
| Brand X                                        |
| Saved Mar 4                                    |
| ---------------------------------------------- |
|                                                |
| Home   Search   Saved   Profile                |
-------------------------------------------------
```

Profile

```
-------------------------------------------------
| Profile                                       |
|                                               |
| user@email.com                                |
|                                               |
| [ Sign Out ]                                  |
| [ Camera Permissions ]                        |
| [ About Confidence Labels ]                   |
| [ Privacy Policy ]                            |
| [ Terms ]                                     |
|                                               |
| Home   Search   Saved   Profile               |
-------------------------------------------------
```

Admin Review Panel

```
-------------------------------------------------
| Admin Review                                  |
| [ Search products or records............... ]  |
|                                               |
| Review Queue                                   |
| -----------------------------------------------
| Source Record 101                             |
| Low-confidence mapping                        |
| [ Open ]                                      |
| -----------------------------------------------
| Product 88                                    |
| Missing carbon metric                         |
| [ Open ]                                      |
| -----------------------------------------------
-------------------------------------------------
```

5. Screen-to-API Mapping

- Home recent items -> GET /api/v1/me/history
- Barcode Scanner -> POST /api/v1/lookup/barcode
- Search -> GET /api/v1/search
- Photo Capture -> POST /api/v1/lookup/image
- Candidate Results -> POST /api/v1/lookup/image
- Result Details -> GET /api/v1/items/{id}
- Source Detail -> GET /api/v1/sources/{id}
- Save Item -> POST /api/v1/me/saved-items
- Saved -> GET /api/v1/me/saved-items
- Remove Saved -> DELETE /api/v1/me/saved-items/{itemId}
- Admin Panel -> admin endpoints

6. Recommended Next Build Artifacts

The next two documents that make this immediately buildable are:

1. API OpenAPI spec
2. Frontend component inventory with navigation and state definitions